

ASTRA DAO

WHITEPAPER

V3.0

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ABSTRACT

Astra DAO is a decentralized and non-custodial automated crypto asset allocator initially built on Ethereum, now migrated to Arbitrum. Astra DAO provides convenient and practical access to crypto-oriented investment strategies. Astra DAO's use case includes providing various investment products/indices, participation units marketplace, user staking, harvesting investment strategies profits, zero-fees user participation model, and an improvement culture using the DAO-managed Treasury. This document describes the tokenomics and theory behind the Astra DAO, explaining the different aspects of the implementation.

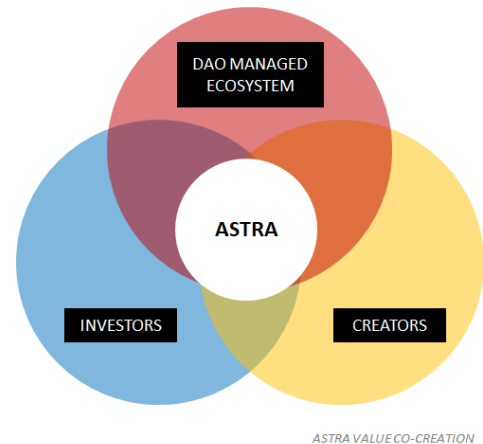
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1. INTRODUCTION

The 2020 COVID-19 pandemic accelerated the digital economy many years into the future and sent the global economy into a recession. With fears of hyperinflation looming, people are looking for more innovative ways to store their wealth, leading to increased portfolio allocations of time-tested stores of value like gold.



With the emergence of blockchain technology and cryptocurrencies, investors are transitioning into a post-COVID, decentralized economy. Bitcoin, the world's most popular cryptocurrency, is commonly referred to as digital gold, with institutions now choosing Bitcoin as a core capital allocation strategy. Bitcoin ushered in a new era of digital currencies by allowing anyone to create a cryptocurrency based on its architecture. Creating new digital assets to solve different problems will become the norm as the world moves toward digital money.

The World Economic Forum (WEF) predicts that in 2027, 10% of the world's gross domestic product, or about \$9 trillion, will be in crypto assets. This fast-moving ecosystem can be volatile, and choosing a suitable investment is equivalent to attempting to find a needle in a haystack. Retail investors need more resources to research crypto assets adequately. Investors may find balancing a diversified portfolio complicated and expensive.

Astra DAO will solve this problem by providing crypto investment strategies to investors and creators. This innovative model creates a win-win incentive for profitable indices investors, creators, and token holders. Index funds have been a popular way for retail investors in the equity market to diversify their investments. Similar to what Vanguard did to Index Funds and ETFs, Astra DAO wants to do for crypto investors. Astra DAO is a technology that will have a mass scale in the new digital economy.

2. VALUE PROPOSITION

Astra DAO's purpose is to provide convenient and practical access to crypto-oriented investment strategies. The core idea of the platform is based on three groups of interest co-creating value – investors (users), product creators (indices/pools creators), and Decentralized Autonomous Organization (DAO) governing the entire Astra DAO ecosystem.

Astra DAO platform will use its ERC-20 native token (“ASTRADA0”) to encourage active participation, facilitate governance, and manage value aggregation and internal cash-flows between users.

The following elements describe the high-level use case behind the Astra DAO platform and its related products:

- **Variety of investment products**

Users (investors) can visit the Astra DAO platform's [webpage](#) and navigate through the list of different products/indices.

- **Participation units marketplace**

After choosing one of the products (index), the user can buy participation units (“*iTokens*”) representing their share in the product.

- **Staking**

Users can also buy and stake ASTRADA0 for additional ASTRADA0 Yield.

- **Harvesting investment strategies profits**

Users can cash out participation units in any index, anytime, simply by depositing them back into the index. In return, they will receive the original deposit plus a proportional share in a profit made by the index (if the index value has increased).

- **Zero-fee user participation model**

No entry fees are collected from users upon buying participation units. An exit fee is collected upon cashing participation units out. The longer users hold the participation unit, the smaller the exit fee gets and ultimately gets to zero.

- **Join & earn**

A portion of the revenue will incentivize index creators to add successful investment strategies
80% of both Performance fees and early exit fees will go to the creator.

- **Redistribution of profits to stakers**

16% of both Performance fees and early exit fees are fully redistributed to the ASTRADA0 stakers.

- **Continuous improvement culture**

DAO-managed Treasury is used to incentivize users, pay interest to ASTRADAO holders, and fund further platform development and continuous improvement.

3. PRODUCT OVERVIEW

3.1. Ecosystem Participants

Important participants in the Astra DAO ecosystem are Stakers, Users, Index/Pool creators, Community members, Contributors & DAO members (governors and voters). The graph shown in Figure 1 summarizes the overall participants' relations model.

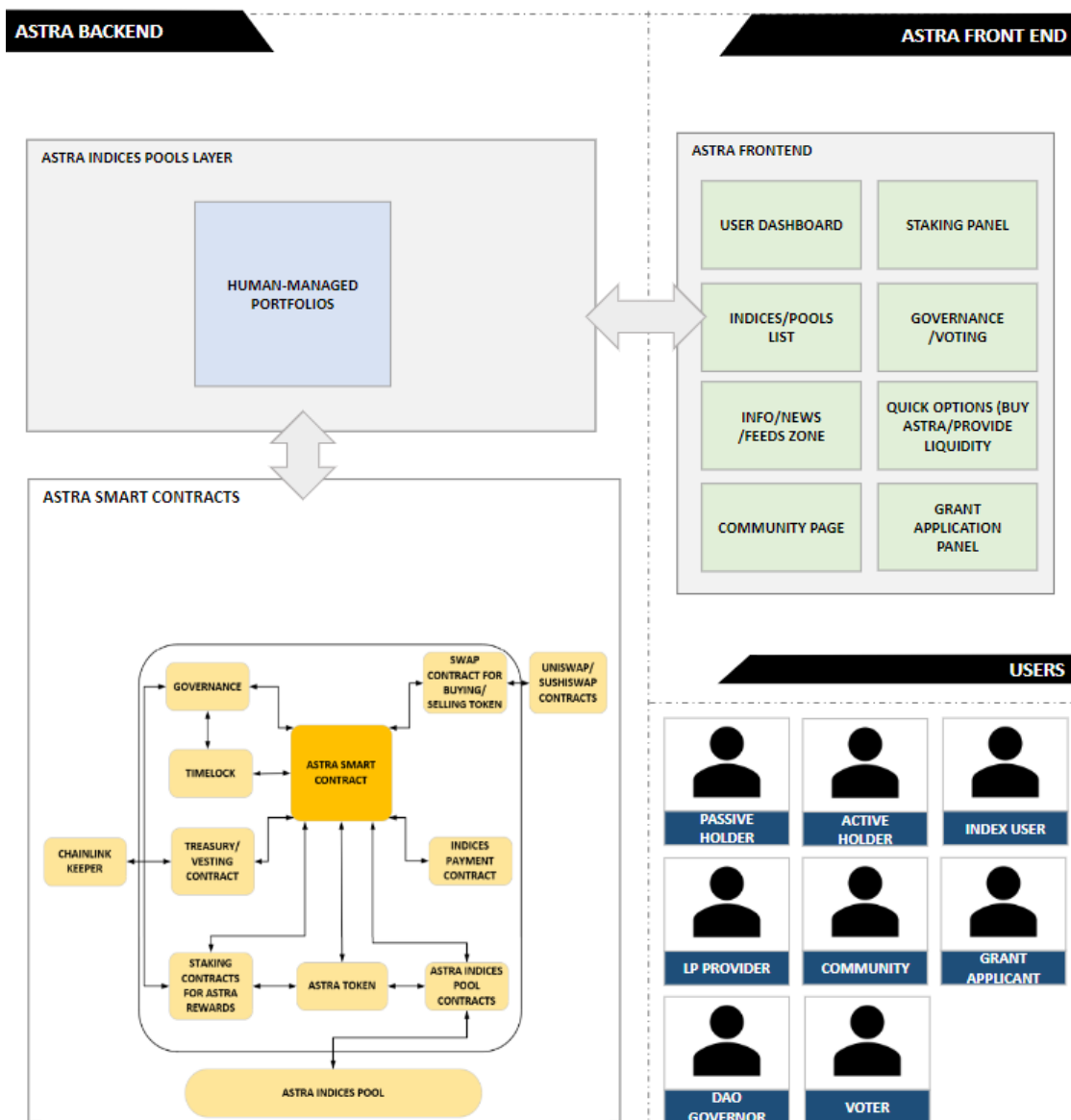


Figure 1: Participants' Relations Model

3.2. Expected Value

Table 1 depicts the expected value analysis by major Astra DAO stakeholders: Investors, Creators (aka "index providers"/"investment managers"), Token Holders, Token Stakers, Other contributors, and DAO.

Table 1 should not be seen as a complete and definite list of all possible stakeholders and values but as the main areas of consideration for core product stakeholders.

Table 1. Expected Value Analysis

Stakeholder	Expected value
Investors	Investors were separated from Token Holders/Stakers due to different Return on Investment (ROI) approaches. The primary and most important expected value is the safety of invested capital, low to medium volatility of assets, a long-term stable increase of token value, and competitive Annual Percentage Yield (APY) for staking/providing liquidity. Investors are more likely to conduct their research on Astra DAO infrastructure and tokenomics. They will focus on understanding and assessing the value for users and the team's core capability to deliver that value. <i>Expected value summary:</i> <i>Asset security, stable competitive APY, transparency.</i>
Creators	Creators (in other words: "investment managers") will see value in the apparent possibility of adding their products/services to the Astra DAO platform. Creators will focus on factors like overall adoption and recognition of Astra DAO platform/ ASTRADA token in the crypto community. These factors will be their points of consideration before choosing Astra DAO as their delivery channel since profits depend on them. We believe the above factors will eventually be the Creators' decisive factor (performance fee is kept at a reasonably competitive level). <i>Expected value summary:</i> <i>Easiness of integration, overall recognition of Astra DAO, total value locked in Astra products, active users counts, community perception, performance fees.</i>
Passive token holders/stakers	Passive holders oppose active holders (stakers using their voting power and utilizing key product components). Generally, passive token holders decrease the overall product traction and dynamic; however, participation in any crypto-related project is inevitable.

	Predicting passive token holders' behavior is challenging, depending on the invested amount and their strategy/maturity. They can play either long or short (the one thing in common is that they expect profit). Holders of unstaked tokens are usually less "attached" to the project. They can be more sensitive to price movements, meaning they are more prone to panic-buy/sell behaviors and vulnerable to so-called "fudding" and "shilling." <i>Expected value summary:</i> Holdings value increase.
Active token stakers	In the Astra DAO context, active token stakers are users interested in the actual usage of everything the product offers. This is the target group (aka "end users") for which the product and its features are designed. It also includes index creators who have to buy and stake ASTRADAO to integrate products with the Astra DAO network or offer other services using Astra DAO channels. <i>Astra DAO end users will be expecting the following value:</i> Competitive ROI generated by Astra DAO indices, a variety of investment products designed to fit their risk appetite and investment approach, low complexity of UI, high-quality UX, the possibility to express their opinion and provide feedback, staking rewards, airdrops, community support, roadmap transparency, continuous project development & improvement, overall project & token value increase, loyalty appreciation.
Other contributors	One of the basic rules of managing product communities and user involvement is rewarding valuable contributions. <i>In this context, the main areas of concern are bug bounties, development grants, improvement/development ideas appreciation, and overall recognition.</i>
DAO	DAO governors and DAO members are responsible for product ownership and setting up the project's overall direction. <i>Areas of interest are effective organizational setup, efficient decision-making model, the minimal administrative effort required, and the possibility to enforce the decisions taken.</i>

3.3. Product Design

Table 2 contains a high-level summary of Astra DAO features in Version One, Two, and Three (v1, v2, v3).

Table 2. High-level Summary of Astra DAO features in Version One, Two, and Three (v1, v2, v3).

v1	v2	v3
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• Adding new custom products by users (creators) • Staking score • Adding human-managed products (investment managers) • Exit fees progressive reduction mechanism • Staking and Lockup vault • Staking cooldown period • iTokens staking • Liquidity mining and Rewards system • Loyalty appreciation (ranks + unlockable benefits) paired with staking score • Governance forum	• Implementation of ecosystem fee • Chainanalysis KYT address blocking of wallets • Gasless Voting • Fix the Stake and Restake functionality issue • Reduction of queue/execution time after a proposal is succeeded • Reduction of staking gas fees • Reduction of staking cooldown period • New slashing fees logic	• Launch on Arbitrum to reduce gas fees and increase efficiency • Cross-chain enabled governance • Bug bounties program
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- Product Usage - Users

Ideally, users (index users, stakers, iTokens stakers, liquidity providers) should be able to use the following features of the Astra DAO platform: Investors dashboard, number of ASTRADAO tokens held/staked, active investments, overall ROI, active votings, APY, unlocked benefits, voting panel, governor panel.

- Product Usage - Creators

Creators (index managers/investment managers) should be able to use the following options: Creator dashboard, number of products added, product ranking, product's total value locked (TVL), ROI, average ROI on all products, holders, user rankings, performance fees earned, performance fees paid.

4. TOKEN UTILITY

4.1. ASTRADAO Token

Utilities Access - Using core products of the platform, such as assets investing, will not require holding ASTRADAO. However, only ASTRADAO holders can vote, add their indices or join the network as investment managers and harvest performance fees.

Mean of Reward - ASTRADAO will be used as a reward for network participants, paid out in staking rewards, liquidity provider (LP) rewards, and structured airdrops.

Development Fund - Part of the Astra DAO treasury pool will be reserved for ongoing development and improvement of the Astra DAO network by funding development grants, bug bounties, and improvement ideas.

Treasury Reserve - Astra DAO treasury will maintain a healthy supply and demand of Astra DAO tokens in circulation. The treasury will execute monetary policy as a long-term strategy to stimulate the healthy growth of the network.

Voting and Voting Power - ASTRADAO holders can participate in voting, submit voting proposals, and act as DAO governors (if they are in the top 100 token holders group). The process is deeply described in the [Governance](#) section of this document.

Staking - Platform participants will benefit from the staking protocol. In the early stage of the development of the network, staking expenditures will be covered using Reward Protocol. As the network develops, staking will be covered with network fees managed by Treasury Pool.

4.2. Index Participation Units (iTokens)

Access to Deposited Funds - Index tokens (also referred to as "iTokens") are non-transferrable ERC20 standard tokens compatible with traditional Ethereum-based wallets. Once received by a user, iTokens cannot be transferred to other wallets and can only be redeemed by being sent back to the index smart contract. All funds deposited into the index smart contract are only accessible to the user who deposited funds.

Participation Shares - iTokens are equivalent to a percentage share in index USDC value. Every profit/loss incurred by the index is transferred to iTokens holders based on their iTokens participation ratio in all iTokens outstanding.

Creators need to pay 5,000,000,000 ASTRADAO to create an index. Once an index is created, the creator or any index participant could create a proposal for the index to become eligible for iToken staking rewards. The process on how to create a proposal is deeply described in the [Governance](#) section of this document. For details on how to stake iTokens, please see the "iTokens staking" section of this document.

4.3. iTokens Distribution

Investors can join any Astra DAO index or investment pool by purchasing iTokens of that index/pool. An index can only be bought with ETH, DAI, USDT, and USDC. The amount of iTokens received is calculated based on the ratio of their allocation value to the entire index value.

Example 1:

- User A puts **500 USDC** in *ABC_index*; he is the first investor; therefore, he receives **500 ABC_iTokens**.
- User B invests an additional **500 USDC** in *ABC_index* and also receives **500 ABC_iTokens**.
- User A has **500 ABC_iTokens** and **50% share** in *ABC_index* value.
- User B has **500 ABC_iTokens** and **50% share** in *ABC_index* value.

Example 2:

- User A puts **100 USDC** in *BCD_index*; he is the first investor; therefore, he receives **100 BCD_iTokens**.
- At some point in time, *BCD_index* doubles its value and is now worth **200 USDC**. User A is still the sole investor and has **100% shares** in *BCD_index*.
- User B is now interested in *BCD_index* and allocates an additional **200 USDC**.
- User B receives **100 BCD_iTokens**.
- User A now has **50% shares** in *BCD_index*, worth **200 USDC**.
- User B has **50% shares** in *BCD_index* worth **200 USDC**.

Example 3:

- User A puts **100 USDC** in *EFG_index*; he is the first investor; therefore, he receives **100 EFG_iTokens**.
- At some point in time, *EFG_index* loses twice its value and is now worth **50 USDC**. User A is still the sole investor and has **100% shares** in *EFG_index*.
- User B is now interested in the *EFG_index* and allocates an additional **200 USDC**.
- User B receives **400 EFG_iTokens**.
- User A now has **20% shares** in *EFG_index*, worth **50 USDC**.
- User B has **80% of shares** in the *EFG_index* worth **200 USDC**.

4.4. Shut-Down an Index

The creator of an index can shut down the index at any given time. Once the creator shuts down an index, the index will no longer accept funds. Investors can withdraw their funds from the index at any time.

5. TOKENOMICS

The total supply of ASTRADA0 is currently capped at 130 Trillion tokens. The supply will only increase or decrease if the DAO votes to implement minting or token burn capability.

ASTRADA0 allocation and token generation event (TGE) unlock chart is shown in Figures 2 and 3.

AstraDAO Token Allocation

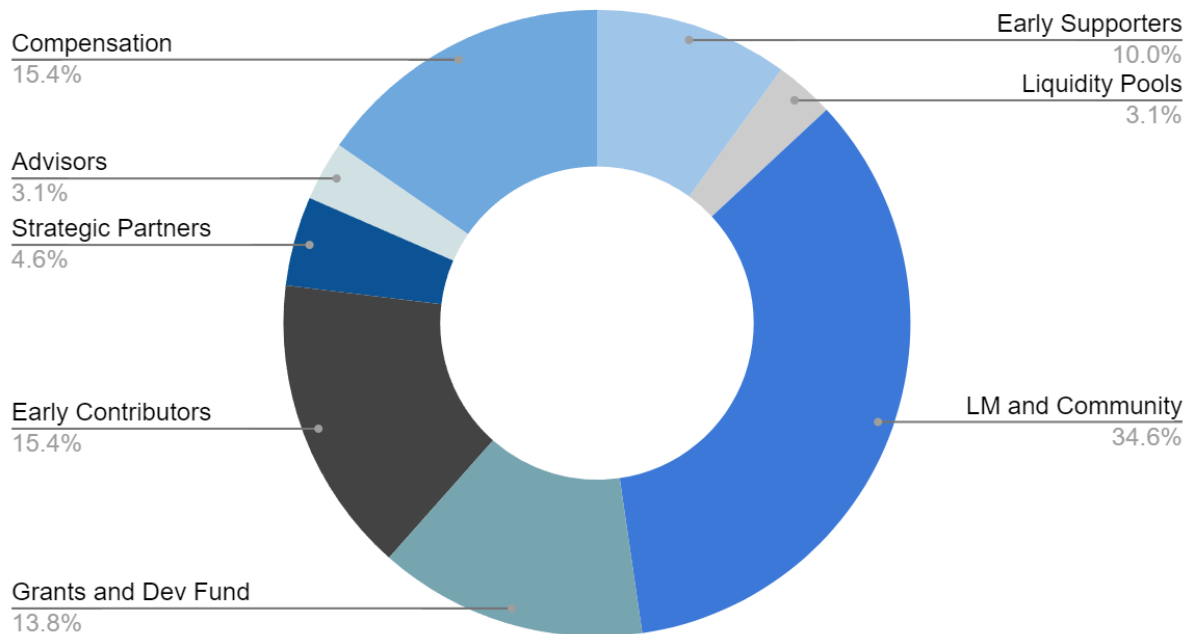


Figure 2: ASTRADA Token Allocation Split

AstraDAO TGE Unlock

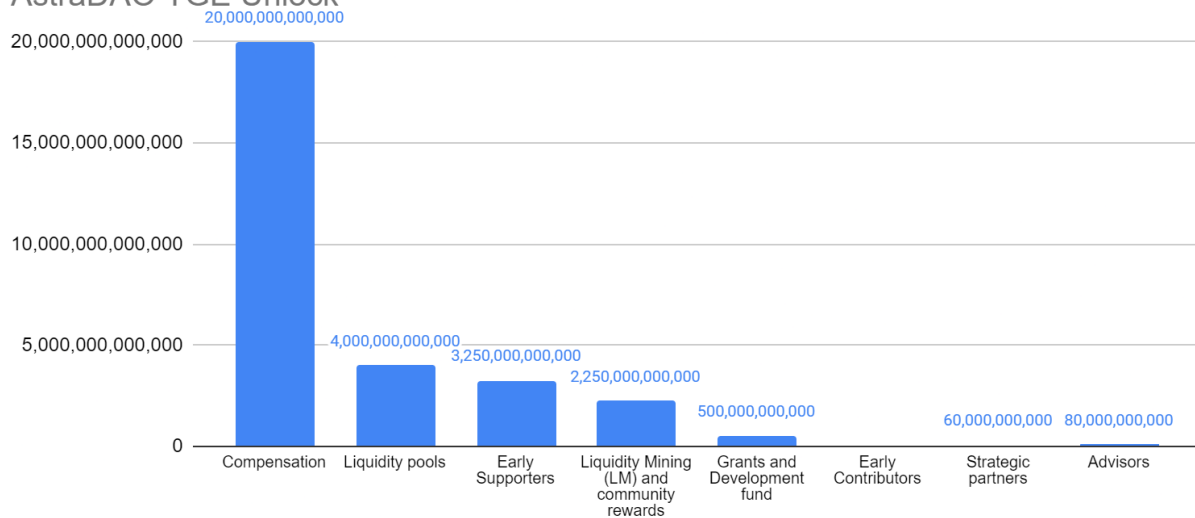


Figure 3: Allocation Split and TGE unlock

5.1. Allocation and Vesting

Table 3 describes the Token ownership allocation and vesting.

Table 3: Token Ownership Allocation & Vesting

Group	AstraDAO allocation	%	Cliff (weeks)	TGE unlock	Vesting (weeks)
Early Supporters	13,000,000,000,000	10%	0	25%	104
Liquidity Pools	4,000,000,000,000	3.1%	0	100%	N/A
Liquidity Mining (LM) and Community Rewards	45,000,000,000,000	34.6%	0	5%	104
Grants and Development Fund	18,000,000,000,000	13.8%	0	5%	104
Early Contributors	20,000,000,000,000	15.4%	12	0%	208
Strategic Partners	6,000,000,000,000	4.6%	0	1%	104
Advisors	4,000,000,000,000	3.1%	0	2%	52
Compensation	20,000,000,000,000	15.4%	0	100%	N/A

The token vesting period is visually shown in Figures 4, 5, 7, and 8.

Weekly Distribution - Global

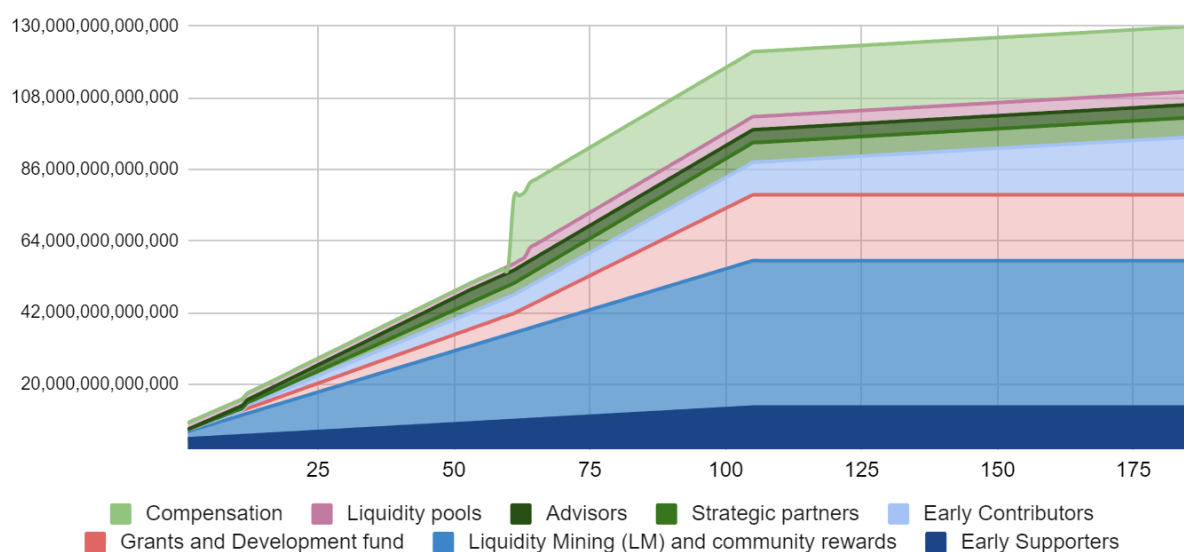
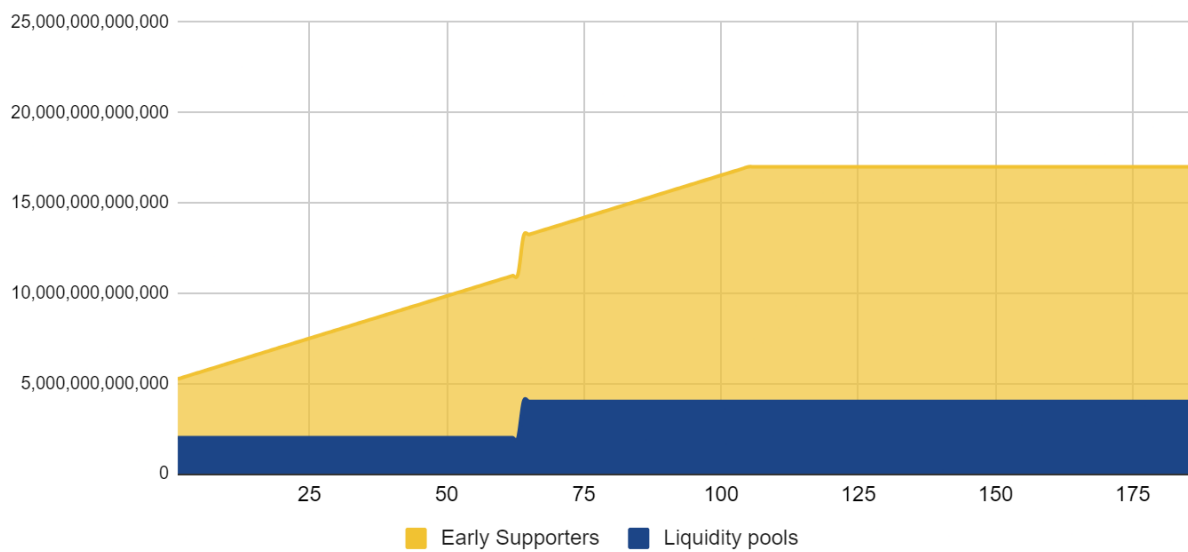
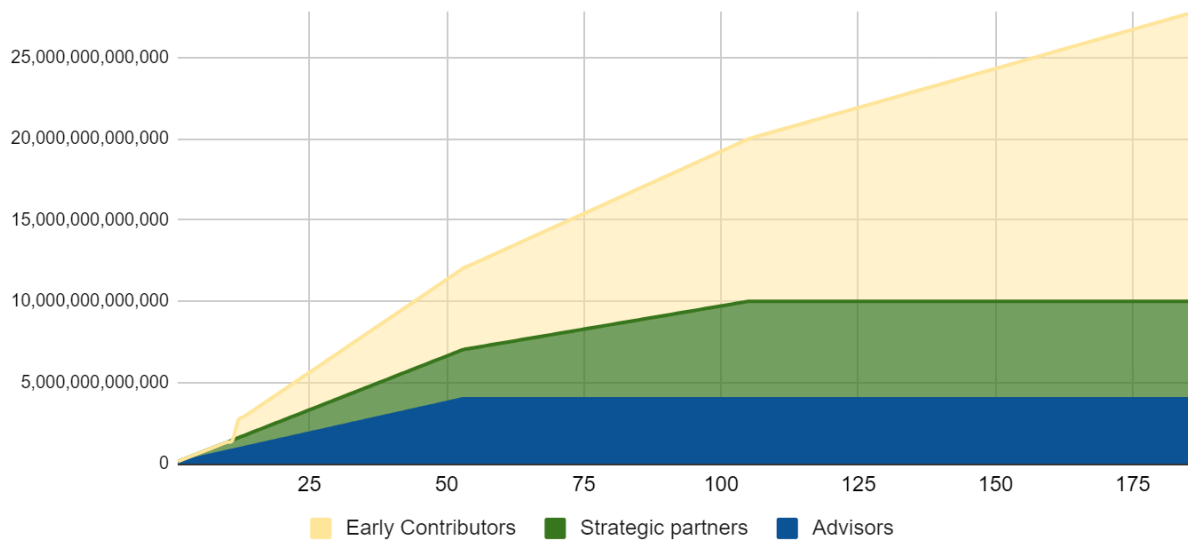


Figure 4: Distribution schedule -Global

Weekly Distribution - Supporters & Liquidity



Weekly Distribution - Internal



Figures 5 & 6: Distribution Schedule for Supporters, Liquidity, Team & Internal

Weekly Distribution - Ecosystem

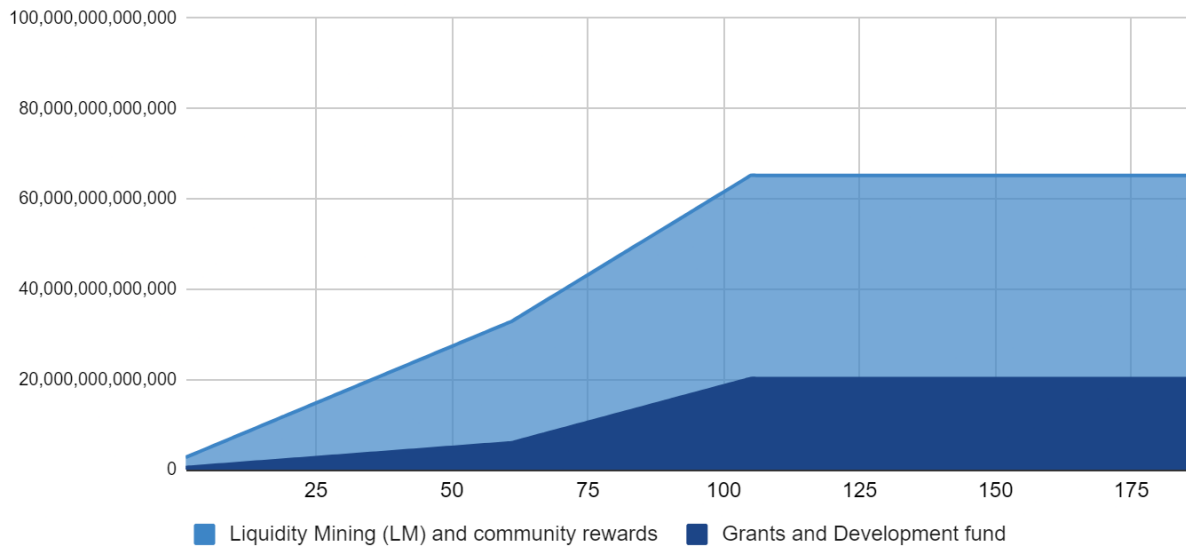


Figure 7: Distribution Schedule - LM and Community Rewards, Grants and Development Fund

5.2. Token Circulation Model/Cash Flow

Astra DAO treasury will be divided into three segments/cash flows (Figure 8):

1. **Redistribution Stream** - 80% of performance and early exit fees go to the index creator as a reward, 16% is converted to ASTRADAO and distributed back to the Astra DAO token stakers, and 4% goes to the treasury.
2. **Rewards Pool** - Created from Astra DAO treasury. Approximately 30% is distributed to Astra DAO token stakers as rewards supplementation, and about 70% is used for iTokens staking/liquidity mining rewards events. It can be refilled if needed.
3. **DAO Managed Treasury** - Reserves and other pools managed by DAO and used for strategic distributions (bug bounties, grants, and others.) Please see the "[token allocation](#)" section for details.

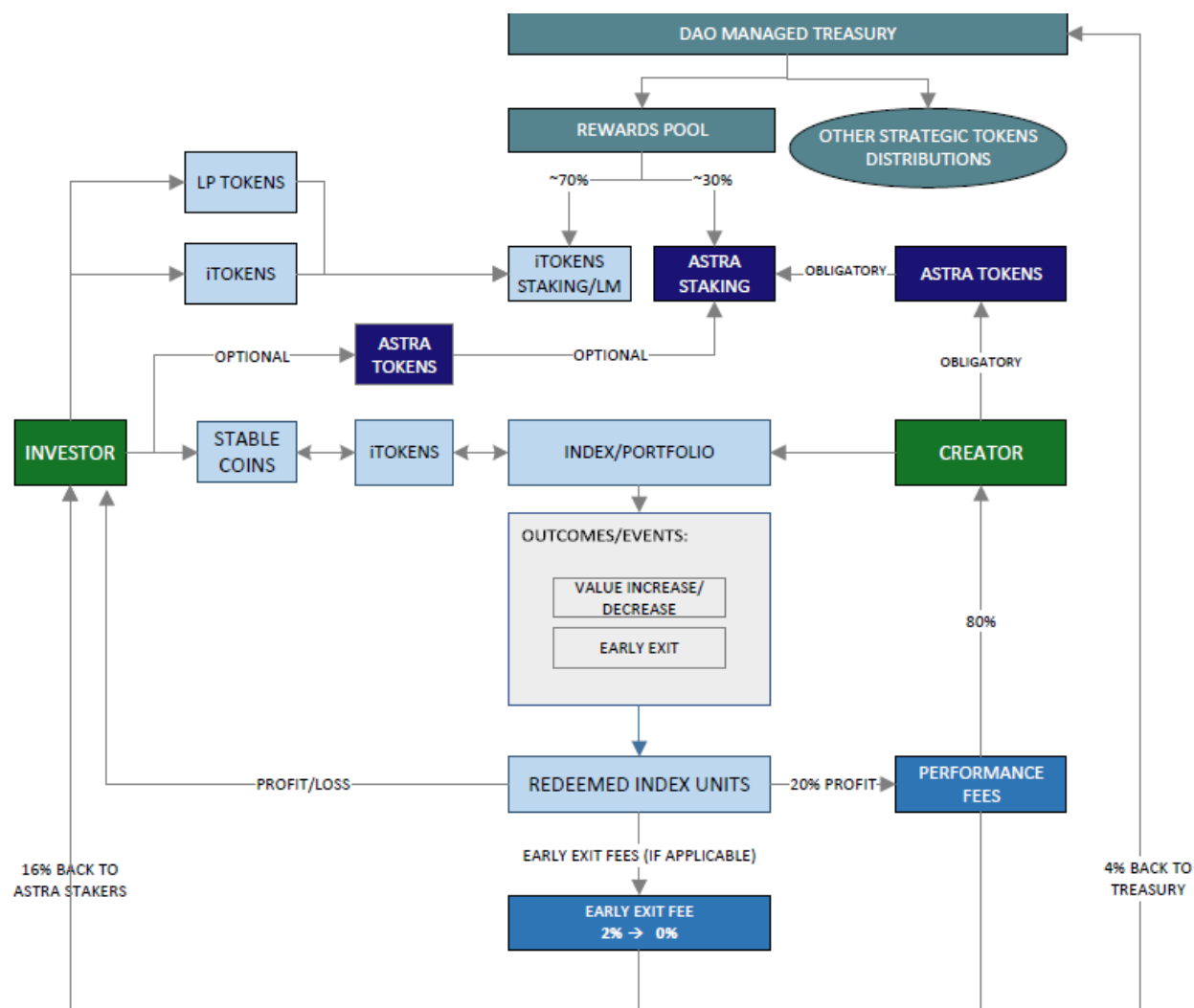


Figure 8: ASTRADAO Token Circulation Model

Astra DAO community will completely control when to pause and unpaue ASTRADAO transfers. This can be done only through a successful proposal created by a DAO community member.

An early exit fee is set to minimize panic/flash transaction behavior and promote longer-term holdings of participation units. (Please see [fees](#) section of this document for more details).

Early exit fee conversion - Instead of paying an early exit fee, investors can opt-in to buy an equivalent amount of ASTRADAO, automatically locked on a staking contract for six months.

5.3. Transactions/Token flow

Figure 9 shows all possible transactions that can take place within the ASTRADA0 ecosystem. Transactions related to rewards claims were described separately in this document's "[Rewards](#)" and "[Rewards distribution system](#)" sections.

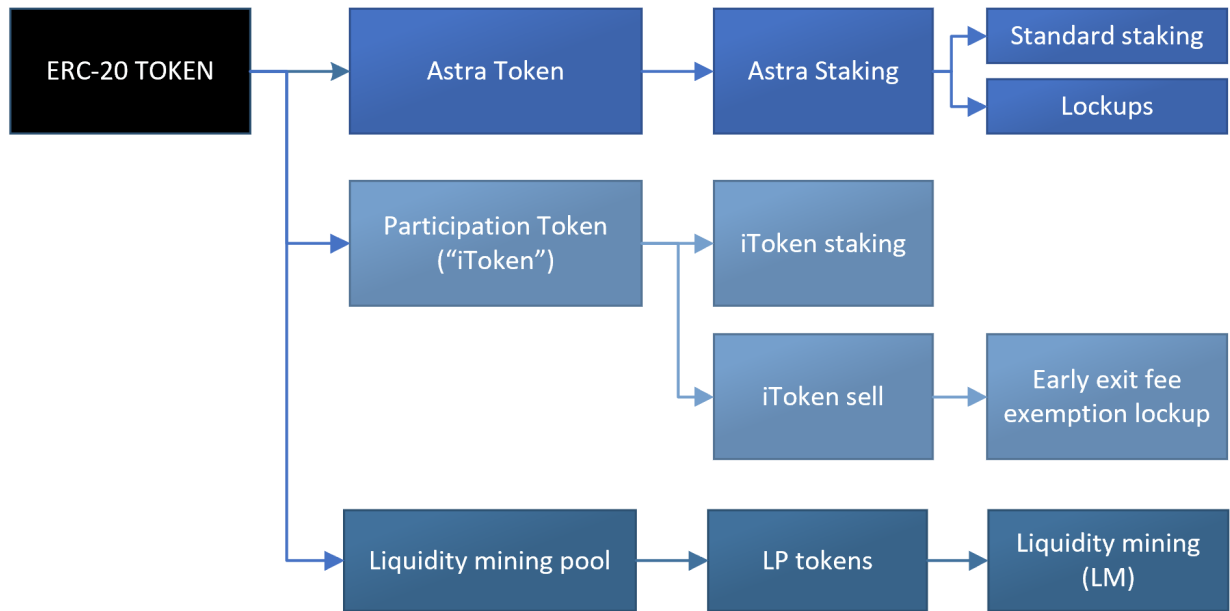


Figure 9: All Possible Transactions Within The ASTRADA0 Ecosystem

The system shown in Figure 9 is embedded into a broader ecosystem, described in the following document sections.

5.4. Astra DAO Staking Model

This section covers staking model solutions and best practices adopted by the Astra DAO network.

Cool-down Period

Stakers willing to withdraw tokens from the staking pool must go through 1 day of cool-down period. Unstake and withdraw after the countdown time elapses. Once assets are withdrawn, the cooldown button will be reset, and users will need to activate the cool-down period again for future withdrawals.

Staking Score

Staking score will promote stakers' loyalty over an extended period. Astra DAO staking score will be calculated as an average of Astra DAO token holdings over the last 60 days.

The staking score aims to recognize the value of a long-term holding even if held assets are small. The below example illustrates this:

- A holder who stakes 1000 tokens for the last 60 days has an average staking score of 1000
- A holder who stakes 60 000 tokens for 1 day also has an average staking score of 1000

Both holders have the same position and would be treated equally in airdrops and rewards distributions. The staking score will be used to identify wallets eligible for airdrops, rewards, and voting.

Holders can always "*stake more*" by adding additional tokens to the staking contract/lockup vault. Staking withdrawal unstakes all tokens (partial withdrawals are impossible) and resets the staking score to 0. The staking score combines the multipliers logic described in the following sections.

Lock-up Period

Instead of transferring to a standard staking vault, Astra DAO tokens can be locked (the staker forfeits the right to unstake them for a fixed period). These are the following lockups vaults: **6, 9, and 12 months**. (Please see the "[loyalty appreciation](#)" section to learn more about the additional benefits of locking AstraDAO tokens).

Multipliers

Depending on users' staking scores and whether they've decided to move AstraDAO tokens to one of the lockups vaults, users will get up to 2.5x higher rewards and voting power (please see "[loyalty appreciation](#)" section for more details).

The rewards related to the staking model were covered in depth in the [Rewards](#) section of this document.

The slashing fee will start from 90% on the 1st day of the Astra DAO staking program and decrease 1% a day over 90 days until achieving 0%. If a user stakes new ASTRADA0 before the 90th day, then the slashing fee will become a weighted average of the old and new ASTRADA0 with respect to the amount and days staked.

5.5. Fees

Table 4 describes the three fees used in the Astra DAO ecosystem: Performance Fee, Astra DAO Ecosystem Fee, and Early Exit Fee.

Table 4: Description of Performance Fee, Astra DAO Ecosystem Fee, and Early Exit Fee

Name	Details
Performance fee	Performance fees are collected automatically upon iTokens redemption if the investor's ROI is above 0. Each deposit made by the investor is tracked separately to detect profit made by that specific deposit (in other words, if the investor bought additional iTokens after 1st deposit, their value is not included in the ROI calculation). If an investor withdraws only a portion of iTokens held, performance fees will be adjusted automatically to the ratio of tokens withdrawn to total tokens held. The performance fee is currently fixed at 20% of the individual investor's profit but can be adjusted by DAO governance vote. Future product versions might give index/pool creators the possibility to define custom performance fees in the 5% to 20% range.
Astra DAO ecosystem fee	Index/pool creators keep 80% of both performance fees and early exit fees, 20% is transferred to Astra DAO treasury as an <i>Astra DAO ecosystem fee</i>. 80% of ecosystem fees are redistributed back to Astra DAO token stakers, while 20% goes to the treasury.
Early exit fee	Early exit fee prevents/discourages investors from redeeming index participation units shortly after buying them (<i>this is unwilling behavior and usually means panic selling, green candle hunting, pump, dump, and other generally unfavorable actions/motivations</i>). The early exit fee is designed to be high at the beginning (2%) and gradually decrease (1/182 of the initial fee daily). After six months of consecutive participation units holding, the exit fee is equal to zero. Users can exempt themselves from early exit fees by converting them to ASTRADAO (with six months lockup).

5.6. Rewards

The following types of token rewards predicted in the Astra DAO ecosystem are:

- Astra DAO staking rewards;
- iTokens staking rewards;
- ETH/ASTRADA0 liquidity mining rewards;
- ASTRADA0 bonuses for redeeming profit in Astra DAO tokens;
- Strategic token distributions used to reward ecosystem participants for valued behaviors (such as but not limited to: *voting, reporting bugs, adding high-performing products, contributing to overall product development, and others.*)

Staking APY

Staking APY is an additional yield gained by users who decide to transfer their tokens to the staking vault. A fixed number of tokens will be distributed per block to the stakers so that the market can find the most suitable equilibrium APY.

iTokens Staking Rewards

iTokens staking rewards are gained by participation unit holders who moved their iTokens on a separate staking vault. For details, please see the "[*iTokens staking*](#)" section of this document.

Liquidity Mining Rewards

Liquidity mining rewards are distributed to ETH/ASTRADA0 liquidity providers on Uniswap or Sushi swap pools. Please see details in the "[*liquidity mining*](#)" section of this document.

Astra DAO Profit Bonus

Investors are provided with two options for selling their participation units:

- **Standard** - their profit equals a proportional share in ROI generated by index/pool.
- **Premium** - their proportional share in ROI generated by index/pool is converted to Astra DAO tokens and locked on staking contracts for additional 6 months. Using the premium option will automatically increase their profit. (please see "[*Loyalty Appreciation*](#)" section for details).

Other Strategic Token Distributions

All strategic token distributions are subjected to DAO governance decisions.

5.7. Rewards Distribution System

Astra DAO ecosystem utilizes two rewards distribution channels:

- **Rewards per block** (from rewards pool).
- **Strategic token distributions** (from DAO managed treasury).

Rewards per block are used as a basic incentivization layer for Astra DAO tokens stakers, iTokens stakers, and liquidity providers by:

- Distribution of tokens to Astra DAO tokens stakers with compounded rewards.
- Distributing Astra DAO tokens to iTokens stakers to increase the indices liquidity depth.
- Distribution of Astra DAO tokens to Uniswap/Sushiswap ETH/ASTRADA0 trading pair.

Strategic tokens distributions are funded directly from DAO managed treasury and include:

- Premium payment bonuses (converting iTokens to locked-up AstraDAO tokens instead of redeeming to stable coin).
- DAO manages other token distributions.

Figure 10 shows the rewards flow and channels.

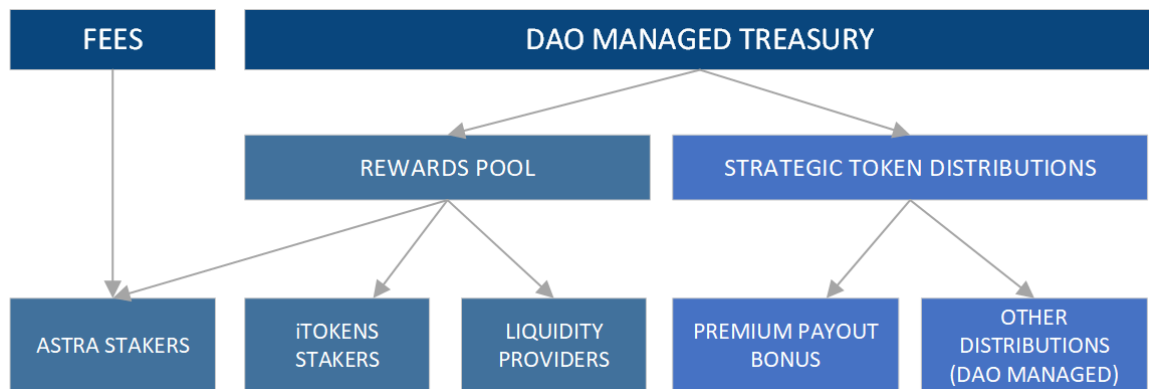


Figure 10: Rewards Flow and Channels

Claiming Rewards

- If you stake Astra DAO tokens, Astra DAO rewards are added on a compounded basis. However, if you stake *iTokens* or *ETH_ASTRADA0_LP_tokens* (uni/sushi), Astra DAO tokens rewards have to be claimed.
- Claiming rewards for iTokens staking and liquidity mining events is subject to *slashing* unless re-staked and locked for a minimum of 6 months.
- Instead of selling, iTokens can be converted to Astra DAO tokens and locked for 90 days. Such conversion provides an investor with premium payout bonus paid in Astra DAO tokens.

Claiming Rewards - iTokens Staking & Liquidity Mining

- The rewards will be displayed to the user like they were available to withdraw.
- Upon claiming the reward, the user is offered two options:
 - Claim 100% and restake in 6 months lockup

- Claim without staking (instant payout)
- If the second option is selected, % of rewards will be slashed, meaning it will be redistributed to the other liquidity miners/iToken stakers pro-rata.
- The slashing fee will start from 90% on the 1st day of the LM/iTokens staking program and decrease 1% a day over 90 days until achieving 0%. If a user stakes new LM/iTokens before the 90th day, then the slashing fee will become a weighted average of the old and new LM/itokens with respect to the amount and days staked.

For example, suppose the user can withdraw 100 ASTRADA0 on the 20th day of the liquidity mining event. In that case, 70% of the rewards are going to be redistributed to other liquidity miners. The user will get 30 ASTRADA0 tokens; 70 will be added to other stakers' claimable amounts.

Loyalty Appreciation - Astra DAO recognizes two dimensions of loyalty appreciation - loyalty measured by **staking score** and loyalty measured by **funds lock-up period**.

Figure 11 and Table 5 explain Astra DAO loyalty appreciation.

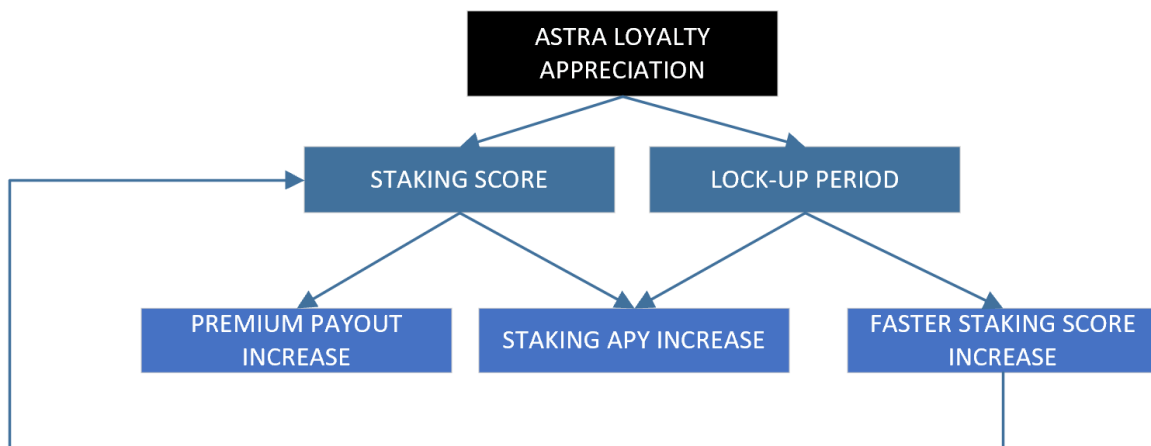


Figure 11: Structure of Astra DAO Loyalty Appreciation

Table 5: Duration and Requirement for the Astra DAO Loyalty Appreciation

Appreciation dimension	Threshold/ requirement	Staking/LM rewards multiplication	Staking score average holding period
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Staking score	100k	x1.2	N/A
	300k	x1.3	N/A
	800k	x1.7	N/A
Lockup period	6 month	x1.1	-20
	9 months	x1.3	-30 (x2 faster-staking score increase)
	12 months	x1.8	-60 (instant full staking score)

You can check details related to the staking score and lock-up vault in the “[Staking model](#)” section of this document. Premium payouts and staking APY are explained in the “[Rewards](#)” section.

Both loyalty appreciation dimensions work together in synergy. Users can increase staking APY using staking score and lockup periods simultaneously (**multiplier will be accumulated**). Moving staked tokens to the lock-up vault will also increase the speed of the staking score increase (by reducing the period used for average holdings calculation). Table 6 shows the multipliers synergy.

Table 6: Multipliers Synergy

		Staking score bonus		
		>=100k	>=300k	>=800k
Lock-up bonus	6 month	x1.3	x1.6	x1.8
	9 months	x1.5	x1.6	x2
	12 months	x2	x2.1	x2.5

Highest possible APY multiplication that can be achieved this way is **x2.5** (80% + 70% = 150%).

Multiplier logic is applied to staking rewards (ASTRADO/iTokens), liquidity mining rewards, and a voting power calculation. The multiplier weights users' participation in rewards sharing to evaluate user participation in the tokens distribution.

An example is shown in Table 7.

There are 100 Astra DAO tokens to be distributed today, and there are three users in the staking contract:

- The first user stakes 100 Astra DAO tokens and does not have a multiplier
- The second user stakes 50 Astra DAO tokens and has a 2x multiplier
- The third user stakes 100 tokens and has a 2x multiplier

Effectively the staking weights of the users are:

- The first user $\Rightarrow 100 \times 1 = 100$
- The second user $\Rightarrow 50 \times 2 = 100$
- The third user $\Rightarrow 100 \times 2 = 200$

Astra DAO rewards token distribution:

- The first user $\Rightarrow 25$
- The second user $\Rightarrow 25$
- The third user $\Rightarrow 50$

Table 7: Example Calculation of Multiplier Synergy

WALLETS				TOTALS
	USER 1	USER 2	USER 3	
Multiplier	x1	x2	x2	
Base capital	100	50	100	250
Multplied capital	100	100	200	400
Share in reward	0.25	0.25	0.5	1
Profit	25	25	50	100

5.8. iTokens Staking

Astra DAO iTokens staking is designed to incentivize participation units ("iTokens") holders. To get iTokens staking rewards, holders must keep their iTokens in an iVault.

iToken holders share rewards based on their % share of iToken holdings and applicable multipliers. Any iToken staking rewards need to be claimed by the user.

NOTE: Proportion is calculated by iTokens amount (please see "[Token utility](#)") section for more details.

DAO will decide the rewards allocation, and essential indices can get more LM rewards than others. In the early stage, iToken will get rewards based on the TVL value in the index, and reward distribution will be dynamic based on the users' share.

The LM incentivization model applied:

1) Flat reward

The reward is distributed evenly to all Astra DAO products. DAO can decide to send more rewards for distribution.

5.9. Liquidity Mining (ETH/ASTRADA0) Rewards

Astra DAO users will also add liquidity directly to DEX liquidity pools (Uniswap/Sushiswap and others) through the Astra DAO [webpage](#). Users will mint **ETH_ASTRADA0_LP** tokens (LP tokens) to calculate their % share in Liquidity Provider rewards.

The logic of LM rewards distribution is the same as in the case of ASTRADA0 staking rewards. DAO can send additional rewards to all the stakers, and they will receive rewards based on their share in the pool. Please see "[rewards distribution system](#)" sections for more details. LP rewards are not auto-compounded and need to be claimed.

5.10. Withdrawal

The withdrawal structure is shown in Figure 12.

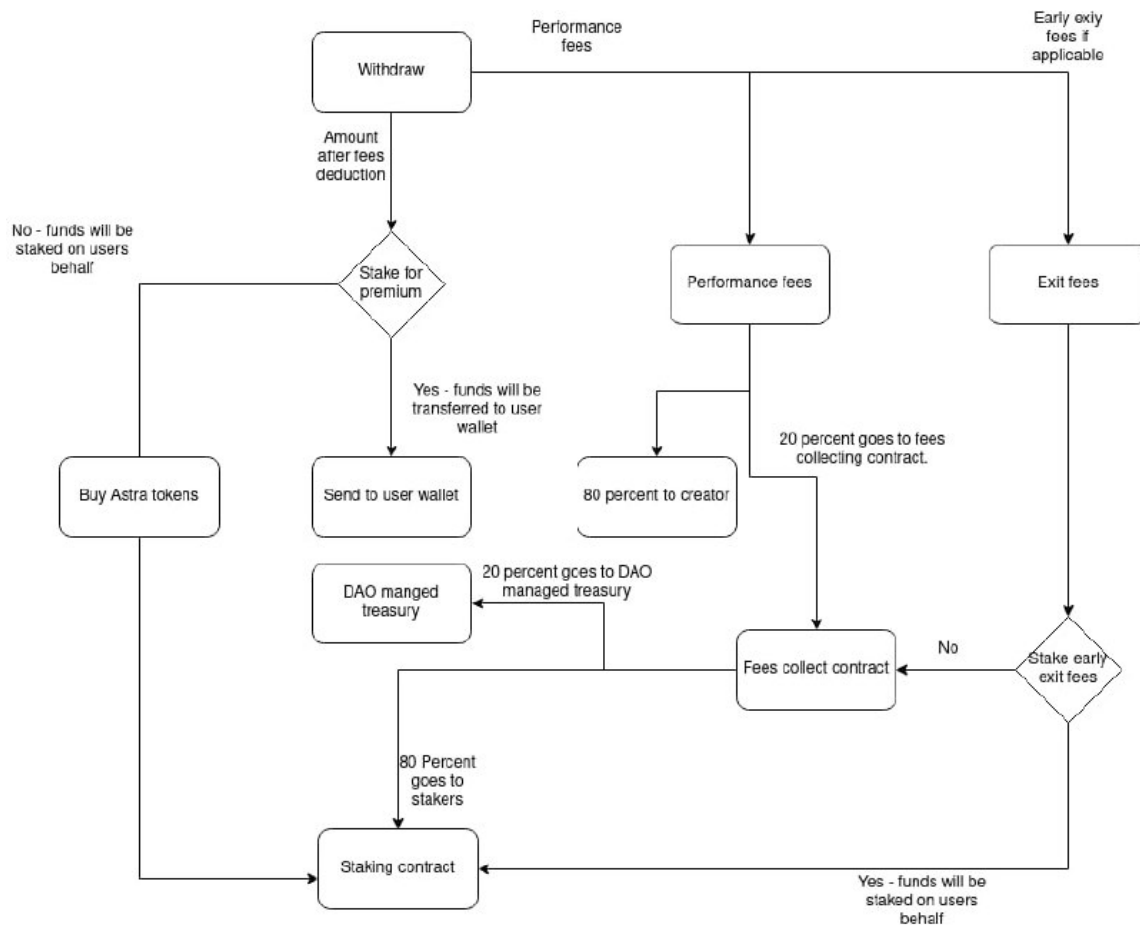


Figure 12: Withdrawal Structure

If a User deposits 100 USDT into the iToken staking vault and on withdrawal, iTokens are worth 200 USDT. The User has made a profit of 100 USDT.

If 20% performance fees are applicable:

- The user gets 180 USDT. 80% of the performance fees will be sent to pool creator, 16% will be distributed among stakers, and 4% will be sent to the treasury. You can find more details in the ["performance fees"](#) section.

If 2% early exit fees are applicable:

- 2% will be deducted from the Users 180 USDT, 80% of the early exit fees will be sent to pool creators, 16% will be distributed among stakers, and 4% will be sent to the treasury. You can find more details in the ["performance fees"](#) section.

- The User has an option of staking to gain higher rewards instead of paying early fees. You can find more details in the "[early exit fees](#)" section.
- After fees deduction, the user can either claim his remaining USDT or stake to achieve a premium bonus. You can find more details in the "[rewards](#)" section.

6. GOVERNANCE

6.1. DAO

During the first 90 days after the Astra DAO network went live, the top 100 wallets performed DAO governance. Currently, anyone can create a proposal and vote.

6.2. Community Grants Program

The Community Grants Program is a decentralized process for the Astra DAO to fund community members interested in furthering Astra DAO's goals and vision. Community members with worthwhile activities that foster the community are welcome to submit proposals for funding. The community has gathered around astradao.org/governance as a website to facilitate discussion, and voting on proposals formally entered on the Astra DAO network.

Proposals may first be publicly discussed as simple pre-proposal posts on any Astra DAO discussion channel, where the community makes feedback and suggestions. Once a proposal owner believes their proposal has a reasonable chance of passing, it may be officially submitted through the Governance Dashboard. A fee of 50,000,000 ASTRADA0 tokens is staked for 6 months to submit a proposal to prevent spam and ensure only important proposals are submitted.

A proposal will include a description of the proposed goals, including details of work to be done and a breakdown of the requested budget. Governors (top 100 token holders) can verify their ability to cast a vote by signing a message from a governor's wallet address. Governors can vote anytime and change their vote until the cutoff time is reached and voting stops. After voting stops, the Astra DAO governance contract executes a decentralized tally and validates all votes. Once a consensus is reached, the results are broadcast, and the budget is allocated. Only Astra token holders may submit grant proposals. Only one proposal can be created per day.

6.3. Voting

Proposals are voted in the order they are submitted. The voting period for each proposal is 7 days. Each Governor can vote once with a yes or no for each proposal. If a proposal passes, DAO Governors will be responsible for its subsequent implementation.

Depending on a matter subject to the voting process, the following requirements must be met for a proposal to pass:

- For any proposals
 - The minimum voting quorum must be at least 33% governors.
 - The number of "YES" votes must be 10% higher than the number of "NO" votes.

If a proposal meets the above criteria within 24 hours of submission, it will be **fast-tracked** and approved in another 24 hours. Astra DAO has implemented a gasless voting mechanism using EIP-712 standard. This means voting will be done off-chain, and all transactions will be submitted to the Ethereum network as a collection. Once voting is completed, any user can initiate and submit the transaction as a collection to the Ethereum network.

6.4. Voting Power

Voting power is expressed in voting points (**VP**). One voting point is equivalent to one staking score point. Staking score multipliers apply to voting power.

Example 1:

- User A bought 500k Astra DAO tokens and locked them in a 6-month vault
- His voting power will increase in the manner shown in Table 8

Table 8: Voting Power Distribution of a User with 500k Astra DAO Tokens Locked on a 6-month Vault

Day	Staking Score	Unlocked multipliers	Voting points	Comment
1	16,667	x1.3	21,667	Users get x1.3 from the start for locking funds on a 6-month lockup vault.
2	33,333	x1.3	43,333	
3	50,000	x1.3	65,000	
4	66,667	x1.3	86,667	

5	83,333	x1.3	108,333	
6	100,000	x1.5	150,000	Users unlock additional x1.2 for Staking score higher or equal to 100k. Accumulated multipliers are now x1.5 (1 + 0.3 + 0.2)
7	116,667	x1.5	175,000	
8	133,333	x1.5	200,000	
9	150,000	x1.5	225,000	
10	166,667	x1.5	250,000	
11	183,333	x1.5	275,000	
12	200,000	x1.5	300,000	
13	216,667	x1.5	325,000	
14	233,333	x1.5	350,000	
15	250,000	x1.5	375,000	
16	266,667	x1.5	400,000	
17	283,333	x1.5	425,000	
18	300,000	x1.6	480,000	Users unlock higher bonuses for Staking score higher or equal to 300k. Accumulated multipliers are now x1.6 (1 + 0.3 + 0.3)
19	316,667	x1.6	506,667	
20	333,333	x1.6	533,333	
21	350,000	x1.6	560,000	
22	366,667	x1.6	586,667	
23	383,333	x1.6	613,333	
24	400,000	x1.6	640,000	
25	416,667	x1.6	666,667	
26	433,333	x1.6	693,333	
27	450,000	x1.6	720,000	
28	466,667	x1.6	746,667	
29	483,333	x1.6	773,333	
30	500,000	x1.6	800,000	
31	500,000	x1.6	800,000	Staking score can't be higher than the staked amount of the tokens. Henceforth, Staking score is the same.

Example 2:

- User B bought 100k Astra DAO tokens and locked them on a 12-month vault
- He instantly gets full possible staking score (x1.3) and an additional bonus for putting funds on a 12-month lockup vault (x1.8)
- His voting power is equal to **210K VP** from day 1 (it can't be higher)

6.5. Community Grants Awards

For Community Grants to pass, the net total yes votes must exceed 10% higher than the number of "NO" votes. If more passing proposals are available than rewards, the proposal that meets all the criteria and executes first will receive the reward. Grant Awards can be nullified at any time if vote totals fall below the approval threshold. Grant Awards are paid in order of positive net votes.

7. TOKEN ECONOMETRICS

This section uses statistical methods, equations, and sample economic data to provide an empirical understanding of network economic relationships and variables. The following items will be subjected to econometrics exercise:

- Distribution of iTokens
- Staking score calculation
- Loyalty multiplier identification
- Distribution of rewards & multipliers logic
 - Staking rewards
 - iTokens rewards
 - Premium payouts
 - LP mining rewards
- Fees calculation
 - Performance fee
 - Early exit fee
 - Slashing fee

USDT was chosen as an example stablecoin. The "USDT" symbol will be used in relevant equations and math.

7.1. Distribution of iTokens

Iv = index value

Pt = total iTokens outstanding

Dv = deposit USDT value

DPv = total USDT value in the pool

$$pTR = iTokens \text{ received}$$

$$\text{If } Iv = 0 \text{ then } pTR = DV$$

$$\text{If } pt > 0 \text{ then } pTR = \frac{Dv}{Iv} \times Pt$$

Examples of scenarios were described in the "[iTokens distribution](#)" section of this document.

The same logic applies to participation in the Astra DAO staking pool and liquidity mining pools.

Table 9: Example of iTokens Distribution

INDEX DEPOSITISTS							INVESTOR ACCOUNTS					
block	event description	User	USDT deposit	iTokens received	iTokens outstanding	Index value	total iTokens held by user A	total iTokens held by user B	User A - value share	User B - value share	User A share value	User B share value
0	index launch					0.0						
1	user A makes 100 USD deposit	A	100	100	100	100	100	0	100%	0%	100	0
2	user B makes 100 USD deposit	B	100	100	200	200	100	100	50%	50%	100	100
3	index value increase x2				200	400	100	100	50%	50%	200	200
4	User A makes 100 USD deposit	A	100	50	250	500	150	100	60%	40%	300	200
5	index value increase x2				250	1000	150	100	60%	40%	600	400
6	User A makes 100 USD deposit	A	100	25	275	1100	175	100	64%	36%	700	400
7	User B makes 100 USD deposit	B	100	25	300	1200	175	125	58%	42%	700	500
8	index value decrease 20%				300	960	175	125	58%	42%	560	400

7.2. Staking Score

The staking score is calculated using average holdings over the last 60 days. As of the time of writing this paper, the average No of ETH blocks per day ("**ABD**") was 6500, meaning that "technically" one 1 "ether day" was equal to the consecutive 6500 ETH blocks processed.

Using the 6500 ABD value, the staking score was calculated using average Astra DAO holdings during the last 390 000 blocks.

ABD = Average number of ETH blocks per day

t = Calculation period

c = Total sum of holdings between block n and n-(ABD*t)

n₁ = Current block

n₂ = Starting block (n-ABD*t)

n = Number of blocks between n1 and n2

Dv = Current deposit value

Ss = Staking score

If t = 0 then

$$Ss = Dv$$

Else

$$Ss = \frac{c}{n}$$

If $Ss > Dv$ then $Ss = DV$

End if

Example 1

- User A makes following deposits to staking vault:
 - +10k Astra DAO tokens in block n
 - +10k Astra DAO tokens in block $[n+20\ 000]$
 - + 300k Astra DAO tokens in in block $[n+300\ 000]$
- User A has not used any staking vault, so approximately the number of previous blocks used to calculate his staking score will be 390 000 (assumed that $ABD = 6500$ than $n = ABD * 60 = 390\ 000$).
- The staking score of user A will increase in the manner shown in Table 10

Table 10: Staking Score Increase for User A (6-month staking vault)

Block No	User staking balance	Deposit Astra	blocks count	Staking score
1	0	10 000	1	0
2	10 000		1	10 000
100	10 000		98	10 000
200	10 000		100	10 000
1 000	10 000		800	10 000
2 000	10 000		1 000	10 000
10 000	10 000		8 000	10 000
20 000	10 000	10 000	10 000	10 000
100 000	20 000		80 000	20 000
200 000	20 000	300 000	100 000	20 000
390 000	320 000		190 000	320 000
600 000	320 000		210 000	320 000
1 200 000	320 000		600 000	320 000

Example 2

- User A has used a 6-month staking vault, so approximately the number of previous blocks used to calculate his staking score will be 195 000 (assumed that $ABD = 6500$, then $n = ABD * 30 = 195\ 000$).
- The staking score of user A will increase in the manner presented in Table 11

Table 11: Staking Score Increase for User B (6-month staking vault)

Block No	User staking balance	Deposit Astra	blocks count	Staking score
1	0	10 000	1	0
2	10 000		1	10 000
100	10 000		98	10 000
200	10 000		100	10 000
1 000	10 000		800	10 000
2 000	10 000		1 000	10 000
10 000	10 000		8 000	10 000
20 000	10 000	10 000	10 000	10 000
100 000	20 000		80 000	20 000
200 000	20 000	300 000	100 000	20 000
390 000	320 000		190 000	320 000
600 000	320 000		210 000	320 000
1 200 000	320 000		600 000	320 000

Example 3

- User A has used a 12-month staking vault. This changes staking score calculation logic and instantly adds a staking score equal to the locked deposit value.
- The staking score of user A will increase in the manner shown in Table 12.

Table 12: Staking Score Increase for User A (12-month staking vault)

Block No	User staking balance	Deposit Astra	blocks count	Staking score
1	0	10 000	1	0
2	10 000		1	10 000
100	10 000		98	10 000
200	10 000		100	10 000
1 000	10 000		800	10 000
2 000	10 000		1 000	10 000
10 000	10 000		8 000	10 000
20 000	10 000	10 000	10 000	10 000
100 000	20 000		80 000	20 000
200 000	20 000	300 000	100 000	20 000
390 000	320 000		190 000	320 000
600 000	320 000		210 000	320 000
1 200 000	320 000		600 000	320 000

7.3. Proper Multiplier Identification

RM = Applicable rewards multiplier

PPB = Premium payout

RM1 = Rewards multiplier unlocked by staking score

RM2 = Rewards multiplier unlocked by staking vault

RM = Total rewards multiplier

RM = $RM_1 + RM_2 - 1$

Tables 13 and 14 are variable tables used for multipliers calculation.

Table 13: Variable Tables Used for Multipliers Calculation (iTokens)

Staking score threshold	Multiplier
100 000	1.2
300 000	1.3
800 000	1.7

Table 14: Variable Tables Used for Multipliers Calculation (month)

Staking score threshold	Multiplier
6	1.1
9	1.3
12	1.8

Example:

User A has a staking score of 350 000. He also locked his funds in a 9-month staking vault.

$$\mathbf{RM} = 1.3 + 1.3 - 1 = 1.6$$

7.4. Distribution of Rewards + Multipliers Logic

The logic behind rewards distributions is the same for Astra DAO staking rewards, iTokens staking rewards and LP mining rewards.

N1 - Start block of pool or last time user claim, deposit, withdraw.

N2 - Current block

$$\mathbf{accAstraPerShare} = \text{accAstraPerShare} + \frac{\text{TotalReward between N1 and N2 block}}{\text{Total token staked}}$$

$$\mathbf{RewardDebt} = \text{accAstraPerShare} * \text{userAmount}$$

$$\mathbf{StakedAmount} = \text{Amount staked by user}$$

$$\mathbf{Reward} = \text{StakedAmount} * \text{accAstraPerShare} - \text{RewardDebt} + \text{UnclaimedReward}$$

Br = Block reward

Bc = User base capital

m = Applicable multiplier

Bm = User multiplied capital

Tm = Total base units outstanding (staked Astra DAO tokens, iTokens, Lp tokens) multiplied.

Cl = Claimable reward

Mtpb (MAX_TOKEN_PER_BLOCK) -This parameter is used to track the maximum reward that can be distributed per block. This Mtpb will be adjusted if users' rewards are slashed in either scenario.

Am -This parameter will be calculated on each deposit/withdrawal to get an average multiplier.

Bl - BLOCK LEFT for reward distribution

Sr - Slashed reward

$$Mtpb = (Mtpb * Bl + Sr) / Bl$$

$$perBlockReward = (Mtpb) / (Am)$$

$$\text{Final Reward per user} = (\text{userStakedAmount} * \text{AccPerShare} - \text{RewardDebt} + \text{Unclaimed}) * \text{UserRewardMultiplier}$$

Table 15: An Example Calculation Showing Claimable Rewards

WALLETS				TOTALS
	USER 1	USER 2	USER 3	
Multiplier	x1.3	x2.5	x1	
Base	100	250	450	800
Multiplier base	130	625	450	1205
% Share in reward	10.79%	51.87%	37.34%	100%
Claimable	54	259	187	500

The detailed examples of rewards distributions scenarios were included in the "[Rewards distribution system](#)" section of this document.

7.5. Performance Fee & Early Exit Fee

PERFORMANCE FEES LOGIC:

x = number of deposits made by an investor

Dep_nValue_{t1} = Value of iTokens bought by [n] deposit, at purchase date [t1]

Dep_nValue_{t2} = Value of iTokens bought by [n] deposit, at withdrawal date [t2]

Dep_nROI = Profit made by deposit [n]

Dep_{tROI} = taxable profit made by all deposits

Dep_{pROI} = Profit already subjected to performance fees paid so far

W = amount of iTokens withdrawn

D = total amount of iTokens held

Perf = performance fees to be paid

Perf_c = performance fees creator's share

Perf_{tr} = performance fees share distributed back to treasury & ecosystem participants.

Method:

Dep_{tROI} = 0

For n = 1 to user deposits count

Dep_n = Dep_nValue_{t1}

Dep_nROI = Dep_nValue_{t2} - Dep_nValue_{t1}

If Dep_nROI <= 0 then Dep_nROI = 0

Dep_{tROI} = Dep_{tROI} + Dep_nROI

Next n

Dep_{tROI} = Dep_{tROI} - DEP_{pROI}

Perf = W/D x Dep_{tROI} x 20%

Perf_c = Perf x 80%

Perf_{tr} = Perf x 20%

Deducting performance fees already paid:

In partial withdrawals, every Dep_nROI value should be lowered by the amount of total Dep_{tROI} already subjected to performance fees so far.

Example: A user made three 100 USDT deposits to index A and received 100, 50, and 25 iTokens. Value increase of those deposits was as follows:

- 1st 100 USDT deposit is now worth 400 USDT (profit = 300 USDT)
- 2nd 100 USDT deposit is now worth 200 USDT (profit = 200 USDT)
- 3rd 100 USDT deposit is still worth the same (profit = 0 USDT)

The investor decided to pull 50 iTokens out and paid 22.86 USD performance fees (400 USDT x 50/175 x 20%). Profit subjected to performance fees was equal to ~114 USD (400 USDT x 50/175). Then Investor decided to pull another portion of 100 iTokens.

The investor will pay a 45.71 USDT Performance fee ([400 USDT – 114 USDT] x 100/125 x 20%). Profit subjected to performance fees was equal to 286 USDT (original 400 minus 114 already subjected to performance fees so far).

EARLY EXIT FEE LOGIC:

EEFr = Early Exit fee rate (Const 2%)

EEFv = Early exit fee value

n1 = Deposit block

n2 = Withdrawal block

n = number of blocks between n1 and n2

ABP = Average block per day (assumed: 6500)

t = early exit fee cool down period (const 182)

Wv = withdrawal value

$$\mathbf{EEFv} = \mathbf{Wv} \times \mathbf{EEFr} - \left(\mathbf{EEFr} \times \frac{n}{\mathbf{ABP} \times t} \right)$$

If **EEFv** <= 0 then **EEFv** = 0

Example:

The following deposit/withdrawal scenario is taking place. Performance fees logic was omitted to maintain the simplicity of the example shown in Table 16.

Table 16: An Example showing a deposit/withdrawal scenario

Block	pToken obtained	Tokens outstanding	Deposit USDT	USDT value in the pool	Event description
1	100	100	100	100	User A deposits 100USDT (first deposit)
2		100		100	
3		100		200	Index value increased x2
4		100		200	
5	100	200	200	400	User B deposits 200 USDT
1 000 000	-100	100	-200	200	User A redeems all pTokens

Calculation :

$$n1 = 1$$

$$n2 = 1\,000\,000$$

$$n = 999\,999$$

$$Np = 200$$

$$t = 182$$

$$ABP = 6500$$

$$EFv = 200 * (2\% - [2\% * 999999/6500/182]) = 0.62$$

User A withdrawal will be subjected to a 0.62 USD early exit fee.

7.6. Slashing fee

SLASHING EXIT FEE LOGIC:

R = Reward amount

T2 = Current timestamp

T1 = Average deposit time

T = Number of seconds in day

Sp = Slashing fees percentage

Sr = Slashed reward

Td = Number of days since average deposit time.

$$Td = (T2 - T1) / 86400$$

$$Sp = \text{If } Td > 90$$

$$Sp = 0$$

Else

$$Sp = 90 - Td$$

$$Sr = R * Sp / 100$$

$$\text{Final reward} = R - Sr$$

Example:

A reward distribution split is shown as an example in Table 17.

Table 17: Reward Distribution Split

WALLETS				TOTALS
	USER 1	USER 2	USER 3	
Multiplier	x1.3	x2.5	x1	
Base	100	250	450	800
Multiplier base	130	625	450	1205
% Share in reward	10.79%	51.87%	37.34%	100%
Claimable	54	259	187	500

- Claim was enabled in block [n]
- User A decided to claim in 51.5 days
- User A will receive 115 Astra DAO token rewards (187 - 72)

The final output for the reward distribution split is shown in Table 18.

Table 18: Final Output

WALLETS				TOTALS
	USER 1	USER 2	USER 3	
Multiplier	x1.3	x2.5	x1	
Base	100	250	450	800
Multiplier base	130	625	450	1205
% Share in reward	10.79%	51.87%	37.34%	100%
Claimable	54	259	187	500
Claimed	0	0	187	
Received	0	0	115	115
Claim increase	21	52	0	72 187

8. CONCLUSION

Astra DAO is a decentralized non-custodial automated cryptocurrency asset allocator platform built on Arbitrium. The Astra DAO network is an innovative model that creates a win-win incentive for index investors, creators, and token holders.

Astra DAO simplifies passive investing for retail participants by providing an array of Crypto Indices. Astra DAO establishes a means for cryptocurrency investing in crossing the chasm and entirely going mainstream.

The most important Astra DAO ecosystem participants are Users, Index/pool creators, Stakers, Community members, Contributors & DAO members (governors and voters).

Astra DAO Token's utility includes its use in development funding, treasury reserve, voting, voting power, staking, adding indices, or joining the network as investment managers, harvesting performance fees, and as a reward for network participants.

Astra DAO allocated 35% of its tokens to LM and community rewards to enable a decentralized ecosystem and provide valuable and appropriate incentives. Astra DAO also allocated 14% to the growth and community grant fund to foster growth and development.

Possible future versions and upcoming iterations will include the possibility of additional rewards for performing products and others.